

: Week-8 (DAY-39) – Handson Keerthivasan R

InsureSure – Microservices-Based Insurance Management System using ASP.NET Core

Problem Statement

Traditional monolithic insurance applications combine policy management, customer onboarding, claims processing, billing, and notifications into a single large system. As the number of users and transactions increases, the system becomes:

- Difficult to maintain
- Hard to scale
- Slow to deploy updates
- Risky (a small change can affect the entire system)

Insurance companies require a scalable, secure, and independently deployable architecture to handle:

- Customer registration
- Policy management
- Claims processing
- Payment handling
- Notifications (Email/SMS)

To solve these challenges, the system must be redesigned using a **Microservices Architecture** in ASP.NET Core.

Objective

Develop a Microservices-based Insurance Management System using **ASP.NET Core Web API**, where each business module runs as an independent service.

Proposed Microservices

1. **Identity Service**
 - JWT Authentication
 - Role-based authorization (Admin, Agent, Customer)
2. **Customer Service**
 - Add / Update / View customer details
 - Maintain customer records

3. **Policy Service**

- Create and manage insurance policies
- Assign policies to customers

4. **Claims Service**

- Submit claims
- Approve / Reject claims

5. **Payment Service**

- Handle premium payments
- Track payment history

6. **Notification Service**

- Send email/SMS notifications
-



Technical Requirements

- ASP.NET Core Web API
 - Entity Framework Core
 - SQL Server (separate DB per service recommended)
 - JWT Authentication
 - API Gateway (optional – e.g., Ocelot)
 - Docker (optional for containerization)
-



Expected Outcome

- Each service can be developed, deployed, and scaled independently.
- Fault isolation (if Claims Service fails, other services continue working).
- Improved performance and scalability.
- Better maintainability and flexibility.

OUTPUT:

 Swagger
version 1.6.0 (beta)

Select a definitionIdentityService v1

IdentityService 1.0 OAS 3.0

<https://localhost:7017/swagger/swagger.json>

Auth

POST /api/Auth/register

Parameters

No parameters

Request body

application/json

```
{  "username": "Keerthivasan_E",  "email": "Keerthi@123",  "password": "Keerthivj@123",  "role": "Admin"}
```

ExecuteClear

Responses

Curl

```
curl -X 'POST' \  https://localhost:7017/api/Auth/register \  -H 'accept: */*' \  -H 'Content-Type: application/json' \  -d '{  "username": "Keerthivasan_E",  "email": "Keerthi@123",  "password": "Keerthivj@123",  "role": "Admin"  }'
```

Request URL

https://localhost:7017/api/Auth/register

Server response

CodeDetails

200

Response body

```
{  "message": "User registered successfully"}
```

Response headers

```
content-type: application/json; charset=utf-8  date: Sun, 15 Apr 2024 13:34:07 GMT  server: Kestrel
```

Responses

Code	Description	Links
200	OK	No links

POST /api/Auth/login

Parameters

No parameters

Request body

application/json

```
{  "username": "Keerthivasan_E",  "password": "Keerthivj@123"}
```

ExecuteClear

POST

/api/Customer

Parameters

No parameters

Request body

application/json

```
{  "firstName": "Harish",  "lastName": "Kalyan",  "email": "harish@gmail.com",  "phone": "777796666",  "address": "128 Main Street, New York, NY 10001",  "dateOfBirth": "2026-04-01T13:41:16.932Z"}}
```

Execute

Clear

Responses

201

Response body

```
{  "id": 16,  "customerId": "CUST2072461",  "firstName": "Harish",  "lastName": "Kalyan",  "email": "harish@gmail.com",  "phone": "777796666",  "address": "128 Main Street, New York, NY 10001",  "dateOfBirth": "2026-04-01T13:41:16.932Z",  "createdAt": "2026-04-19T13:44:09.0417316Z"}}
```

Response headers

```
content-type: application/json; charset=utf-8  date: Sun, 19 Apr 2026 13:44:09 GMT  location: https://localhost:7070/api/Customer/16  server: Kestrel
```

Responses

Code	Description	Links
200	OK	No links

GET

/api/Customer/{id}

Parameters

Cancel

Name

Description

Execute

Clear

Responses

200

Response body

```
{  "id": 1,  "customerId": "CUST007938C",  "firstName": "Harish",  "lastName": "Kalyan",  "email": "harish@gmail.com",  "phone": "777796666",  "address": "128 Main Street, New York, NY 10001",  "dateOfBirth": "2026-04-01T13:41:16.932Z",  "createdAt": "2026-04-19T13:42:25.2857596"}}
```

Download

[illegible]

DELETE /api/Customer/{id}

Parameters

Name	Description
id * required Integer(32x12) (path)	9

Execute Clear

Responses

```
Curl
curl -X 'DELETE' \
'https://localhost:7076/api/Customer/9' \
-H 'accept: */*' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpvcC99eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1NiIsImNpb2kiOiJkaWYiLCJpcyB1bmkiOiJkaWYiLCJwcm9kdGVuaWUiOiJkaWYiLCJhdXRoIjoiZm9udC99'`
```

Request URL:

https://localhost:7076/api/Customer/9

Server response

Code	Details
200	Response body

```
{
  "message": "Customer deleted successfully"
}
```

Download

Response headers

```
content-type: application/json; charset=utf-8
date: Sun, 19 Apr 2025 13:43:54 GMT
server: Kestrel
```


Download

Also, there is

/api/Payment/customer/{customerId}

Parameters

Name	Description
customerid * required integer(5 or 6) (path)	8

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
'https://localhost:7061/api/Payment/customer/8' \
-H 'accept: */*' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyTeOjFudHk2NnQ0ZmVlY2EwLWpYMDM1LTBxMTU0dDkzMDE3eHh0cygkeH02L2pvc3R5cm9kaS9FaWtlL1LLC3o0PDU0LScuOnBuZGVycy9hdXRsZWFrLnphbnQ0' 
```

Request URL

```
https://localhost:7061/api/Payment/customer/8
```

Server response

Code	Details
200	Response body<pre>{ "id": 3, "paymentId": "MY28359873", "customerId": 8, "policyId": 1, "policyNumber": "RD.4F9LS242", "amount": 1000, "paymentMethod": "Credit Card", "transactionId": "TOD8269419143K25", "status": "Completed", }</pre>

Responses

Code	Description	Links
200	OK	No links

[illegible]

$\wedge$

Cancel

Close

10

1

Download

Links

No links

Cancer

1

```
application/json
```

Clear

1

1

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Download

1

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